Nuclear Station IoT Monitoring

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Abstract

In today's rapidly evolving world, improving the scientific and technological environment in Egypt is the top priority Grand challenge facing the nation. To integrate scientific research with industry interests, the chosen solution for this challenge aims to **automize monitoring systems** in nuclear reactors, focusing on crucial factors of the reactor like core temperature, the flow rate of coolant water and radiation, as well as **storing** and **visualizing real-time data** of the reactor on a **user-friendly dashboard**. The monitoring system is to make an **alert** in case the core gets close to the forging temperature with a margin of less than **25 °C**, detect if any UV radiation is leaked, as well as update the dashboard once every **10 seconds** at most. Although it went through a negative result in the beginning due to an error in the code, that tackled the UV detection, the solution was deemed successful after fixing the problem, as our system has successfully achieved all design requirements. The data of all parameters was collected, saved to a cloud database, and visualized on the dashboard with an update every **8 seconds** of their current state. Additionally, an **alert** was triggered successfully when the core temperature reached **227 °C**, as the core is unstable at such temperature, approaching the **forging temperature** with a margin of only **23 °C**. Finally, when the UV lamp was activated, radiation was detected with an average wavelength of **255 nm**. The conclusion drawn from the results and their analysis indicates that using flowrate to detect the pump operability was an efficient and cheaper method, and there is an inverse correlation between the rate of change of temperature and coolant water flow rate.

Introduction

To support Egypt's position as a player in modern economics and politics, the grand challenges facing Egypt must be overcome. One to be focused on is improving the scientific and technological environment in the country. Since there's limited integration between academic research and industry, research is rarely commercialized, and investors aren't keen on funding research applications & technology, as only **1.02%** of Egypt's GDP is spent on research and development ***(Research and development expenditure (% of GDP), 2024)***. Specifically, the targeted problem is **improving the technology of monitoring systems in the new nuclear energy systems**, as they are the current trend for investors. According to research conducted by the University of Texas on 10 nuclear energy centers, **80%** of incidents recorded were the result of either a worker's error or a poorly designed procedure ***(Bohigas & De Fortuny, 2014)***. Automating the system can reduce those human-made errors.

There have been efforts to solve this problem in the past. For instance, the **Olkiluoto Nuclear Power Plant**, which installed a 3rd unit in it as recently as 2021, is known for its **IoT-based monitoring system**. It's divided into multiple systems, each to monitor a specific section in the reactor. It measures most elements directly, for example, to check if the pump in the system is working, it would measure its vibration continuously ***(Rostedt & Koivunen, 2021)***. This while having its merits, still has the problem of being very costly. Another nuclear plant monitoring system was the one established in **Fukushima Daiichi**. When a tsunami hit the reactor, it disabled the cooling system, which led to an explosion and the release of nuclear waste into the air ***(World Nuclear Association, 2024)***. The problem was that the breakdown went **unnoticed** until it was too late in addition to not taking account of the natural conditions of the area.

Before choosing a solution, design requirements have been established that must be achieved by the solution to be deemed a success. First, the system should display information that reflects the real-time state of the station, to act quickly in case any change or unexpected error occurs. It's required for the displayed data to update every **10 secs.** at most. Second, it should have an **alert** that triggers when a problem is predicted to occur, specifically regarding the melting of the core, by keeping a margin of 25ºC from the forging temperature. Third, it should detect any **radiation emitted**, to avoid any harm caused by long exposure.

The solution was chosen to meet these requirements. It will monitor the state of three factors. The nuclear core temperature, as control rods that limit the reaction speed can be added or removed depending on it, **the coolant water flow rate** indicates the state of the cooling system, as this is cheaper than measuring the pump vibration directly, and **UVC radiation leakage**, for being one of the most common radiations in nuclear reactors. The data collected by the sensors was stored in a **cloud database** because it provides more scalability and accessibility from any device, as well as providing a backup of the data and updating immediately. The data from the temperature sensor was analyzed and compared with the forging temperature of the core. All this data will be **visualized** on the online visualizing tool, Adafruit IO, to make the **data user-friendly**. A prototype was built on a simulation of the reactor, and the results were a success.

Materials & Methods

Table (1): Materials Table with details

Item	Amount	Description	Cost	Source	Picture	Item	Amount	Description	Cost	Source	Picture
Gas Heater	×1	Fire igniter uses natural gas	40 L.E.	Gas Cylinders Supplier		Breadboard	×1	Contact board for creating electronic circuits	27 L.E.	Electronics Store	
Heating Container	×1	5L Aluminum Container	60 L.E.	Kitchenware Store		YF-S401	×1	Water Flow Rate Sensor	250 L.E.	Electronics Store	
Water Container	×1	6L Plastic Container	5 L.E.	Plastics Store		Thermocouple MAX-6675 with k-type sensor	×1	Withstand extreme temperature	225 L.E.	Electronics Store	
Hxchen R385 Pump	×1	12V liquids pump	230 L.E.	Electronics Store		GUVA-S12SD	×1	UV light sensor	120 L.E.	Electronics Store	
Potentiometer	×1	Three terminal resistor with a rotating contact	5 L.E.	Electronics Store		Jumper Wires	×18	Copper wires covered by insulating Plastic	0.75 L.E.	Electronics Store	
Hose	1.20 m	120 cm rubber hose of 0.5 cm diameter	20 L.E.	Plumbing tools Store		Buzzer	×1	Electric to Sound signals converter	7.5 L.E.	Electronics Store	
UV Lamp	×1	A 230V UV Lamp with transformer	350 L.E.	Electronics Store		Batteries	×3	3.7V Lithium Ion	50 L.E.	Electronics Store	
ESP32	×1	Microcontroller with built in Wi-Fi module	350 L.E.	Electronics Store		Battery Case Holders	×2	2 Cells Holder	15 L.E.	Electronics Store	
Total Cost: 1,883 L.E											

Methods:

To start the journey of the construction of the prototype using the materials mentioned in **Table (1)**. The process is divided into two parts: Simulating variables monitored from the nuclear power station, and installing the monitoring system.

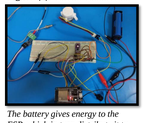
As illustrated in **Figure (1)**, the 120 cm hose is cut into three parts of lengths 20cm, 60cm, and 40cm for (container to pump), (pump to water flow sensor), and (water flow sensor to container) respectively. A hole of diameter 0.5 cm equal to the diameter of the hose is made in the water container using a sharp cutter. The 20cm hose is adhered to the hole in the container using wax. The other end of the 20 cm hose is inserted into the inlet nozzle of the water pump. From the outlet nozzle of the pump, the 60 cm hose is inserted from one of its ends, and the other end is inserted in the inlet nozzle of the water flow sensor. For the outlet of the flow sensor, the 40 cm hose is inserted. All the nozzles of the pump, the flow sensor, and even the hole of the container are surrounded by insulating thread seal tape to prevent any water leakage. The water flow sensor is then placed vertically on a piece of wood to pour water into the heating aluminum container placed over the gas heater. Using heat-resistant tape, the thermocouple is attached to the outer surface of the heating container, and the UV lamp is tapped to the wooden column to be connected to electricity.

Figure (1): Water Circulation

The figure shows the flow of water from source container, controlled by pump, to the simulated core

The second part of the journey of prototype construction is connecting the sensors to the ESP. Two wires are connecting between the 3.3V and GND pins of the ESP and the breadboard to be common power pins for all the components. Each component's GND and VCC pins are then connected by two wires to the GND and 3.3V power pins in the breadboard in parallel. The Signal Output (SO), Chip select (CS), and Serial Clock (SCK) pins of the thermocouple are connected to ESP pins: D18, D19, and D21 respectively. The SO pins of the water flow sensor (yellow wire) and UV light sensor are connected to ESP pins D4 and D25 respectively. The buzzer negative leg is connected to the GND of the breadboard while the positive leg is connected to the ESP D13 pin. The code for controlling the system was written in C++ language using Arduino IDE, a software app, including functions to connect the ESP to Wi-Fi, upload to dashboard, and upload to the Google Sheets.
The code is then uploaded to the ESP32 Board using a Micro USB Data Syncing cable. After the code is uploaded successfully, the ESP is disconnected from the laptop and connected to batteries for power. Two 3.7V batteries are connected in series in their cells in the battery holder and the positive (red) wire of the holder is connected to the ESP VIN pin and the negative (black) wire is connected to another ESP GND pin. The Circuit connection is shown in **Figure (2).**

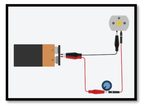
Figure (2): Circuit Connection



The battery gives energy to the ESP, which in turn distribute it to each sensor connected to it .

For powering the Water Pump, the 12V pump is powered using three 3.7V batteries and to control the speed of output water the power of the pump is controlled using a potentiometer. The positive wire of the batteries' holder is connected to the middle leg of the potentiometer and one of the lateral legs is connected to the positive pole of the pump, while the negative pole of the pump is connected directly to the negative wire of the batteries' holder as shown in **Figure (3).**

Figure (3): Water Pump Connection



The potentiometer connects between the positive poles of the battery and the pump, controlling the pump's input voltage.

Test Plan:

To test the success of the prototype, the design requirements are tested in three steps. Face masks and safety goggles should be worn for protection.

Firstly, the UVC radiation detection is tested using a UV Lamp and the wavelength readings are compared to the UVC wavelength range (< 280 nm). Secondly, the core temperature is increased by increasing the intensity of the gas heater, approaching the forging temperature to test the reaction of the alerting buzzer indicating the problem. Finally, testing the time within changes in variables updates on the dashboard. The water flow rate is changed by changing the voltage of the water pump by the potentiometer, and with every change in voltage, the time the change in water flow rate updates on the dashboard is measured using a stopwatch.

Results

We conducted tests in two different trials, one positive and one negative. The results of each trial were collected and shown below:

Negative Trials:

When GUVA-S12SD UV Sensor detected UVC, the output readings of wavelength wasn't in the known wavelength range of UVC (100-280 nm), while when not exposed to UV light the readings of wavelength indicated presence of UVC contradicting the design requirements as shown in table (2).

Table 2: Negative Trial of UVC Detection

UV Lamp State	1st Reading	2nd Reading	3rd Reading	Avg. Wavelength Reading
On	355 nm	366 nm	359 nm	360 nm
Off	246 nm	248 nm	253 nm	249 nm

Positive Trials:

The detected UVC wavelength reading lied in acceptable range achieving the design requirement as shown in table (3). Time response of data update on dashboard for any change in monitored factors was less than 5 seconds and the alerting system responded with a margin from forging temperature less than 25°C achieving all design requirements as shown in tables (4), (5).

Table 3: Positive Trial of UVC Detection

UV Lamp State	1st Reading	2nd Reading	3rd Reading	Avg. Wavelength Reading
On	265 nm	252 nm	248 nm	255 nm
Off	352 nm	359 nm	357 nm	356 nm

Table 4: Time Response of Data Update on Dashboard

Trials	Operating Voltage	Water Flow Rate	Time Response
1st Trial	6V	2.5 L/min	8 secs
2nd Trial	9V	4.3 L/min	9 secs
3rd Trial	12V	5.8 L/min	8 secs
Average Time Response = 8.3 secs			

Table 5: Alert Response to Temperature Increase

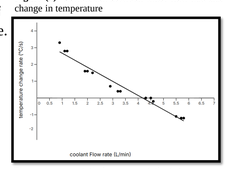
Core Temperature Reading	120 °C	184 °C	213 °C	227 °C	231 °C
Temperature Margin	130 °C	66 °C	37 °C	23 °C	19 °C
Alert Response	Off	Off	Off	On	On

Analysis

Improving Egypt's scientific and technological environment is a critical issue that affects the country's overall development in all sectors. In particular, the lack of modern methods for monitoring energy production is a significant challenge facing the energy sector. As Egypt turned to Nuclear Power as a modern energy source, enhancing the efficiency and safety of Nuclear plants is a crucial concern. A significant challenge in Nuclear plants is the extreme dependency on human power, which threatens the plant's efficiency and safety. To address this problem, an automated monitoring system is needed.

The system will track the critical parameters that hold the plant's production efficiency and ensure human safety. Those parameters are nuclear core temperature, coolant water flow rate, and UV excess radiations. First, monitoring the temperature of the nuclear reactor is crucial as it directly controls the rate of water turning to steam in advance of driving the turbines, determining the reactor's overall efficiency. In addition, when the core temperature approaches its limit, the forging temperature, indicates the possibility of a disaster. Next, the coolant water flow rate should be closely traced to ensure the core temperature doesn't exceed the threshold safety value, avoiding sudden eruptions in the nuclear core. Coolant water affects the core temperature directly as water flow rate is inversely correlated with the rate of change of temperature, the more amounts of water, the more the heat absorption, and the lower the core temperature. Using scatter plotting, a line of regression was drawn in the form of Y= mX + b to predict the rate of the change from the flowrate, as shown in **Figure (4)**. The slope(m) and y-intercept(b) of the regression line was calculated to be -0.84 and 3.5 respectively using the following equation:

Figure (4): Correlation between water flow rate and change in temperature



The graph represents the inverse correlation between water flow rate and core temperature

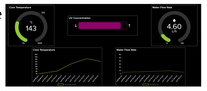
$$m = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2} \quad , \quad b = \frac{\sum y - m\sum x}{n}$$

(ST.3.02; (Upton & Cook, 1996)).

Further, excess UV radiation causes extreme human fatigue and indicates system deficiencies; so, detecting radiation leakage is essential in Nuclear plants. Finally, visualizing the collected data is the cornerstone for taking control of the monitoring process and enhancing monitoring efficiency, allowing authorities to take a life overview of the plant.

For this purpose, electronic sensors were used to detect each parameter minimizing the likelihood of human errors, while IoT techniques were used for data visualization, as shown in **Figure (5)**, including using Wi-Fi protocols for wireless data transfer **(PH.3.04)**. Google spreadsheets were used for data storage due to its cloud-based capabilities which enhance data accessibility, storing, and updating in real-time **(CS.1.05)**. When it comes to sensors, the thermocouple temperature module K-type chosen rather than S, N, and R types, the K type is the only module detecting temperatures within our desired range, 0 °C and 800 °C. Regarding water flow rate, water flow module YF-S401 was the best choice due to its flow rate range,1-30 L/min, aligning with its accuracy (±10%), in addition to its efficient mechanism, which is driven by moment of inertia about the center **(ME.3.01)**. For UV, the GUVA-S12SD UV sensor module was preferred over the GY-8511, because its detection range of 240 nm to 370 nm includes UVC radiation, which is the most prevalent type produced in nuclear plants. To enhance the durability of the system, lithium-ion batteries are used as a power source, with a lifespan of minimum 6 months. To chose appropriate batteries, the total capacity needed by the system was calculated through the law C = Power × time. With a voltage of 3.3V, and total consumed current(I) of 271 mA, the consumed Power was calculated to be 0.89 Watt, P =I × V; so, the total capacity needed is calculated to be 3863.4 Watt-hours.

Figure (5): Data Visualization



The figure shows the dashboard visualizing the monitoring system sensors' readings

Following the construction of the prototype, the UVC presence, updating the dashboard within a maximum of 10 seconds, and the alert for the forging temperature were tested. UV lamp was used to test the detection of UVC. At first, irrational readings were present. When the UV light was turned off, the readings indicated a wavelength of 240 instead of 370, which indicates UVC presence. Conversely, when the lamp was turned on, a wavelength of 370 was indicated instead of 240, which indicates UVC absence. This issue stemmed from a technical issue in the ESP32 code, specifically with the use of the map() function. The map function written in C++ is used to re-map a variable from one range to another. It takes five values: the variable to be mapped, the 2 limits of the original range, and the 2 limits of the new desired range. The problem arose from a misunderstanding of the UV wavelength ranges. In the light spectrum, as the frequency (F) increase, the wavelength (λ) decrease, according to the law:

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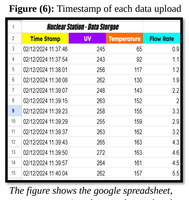
$$C = \lambda F$$

where C is the speed of light ***(Serway et al., 2016)***. The UV light ranges between 100 to 370 nanometers (nm) **(PH.3.02)**, further divided into UVA (315-400 nm), UVB (315-280 nm), and UVC (280-100 nm) ***(Ultraviolet (UV) Radiation, 2017)***. The UV sensor's electrical pulse readings, which range from 0 to 4095, were converted to a corresponding wavelength. However, mistakenly the output range was reversed, mapping the input to an output range of 240-370 nm instead of 370-240, which resulted in indicating UVC presence when the lamp was off. After that, the dashboard update was tested to ensure the design requirement is achieved, as data should be updated within a maximum of 10 seconds. The power supplied to the water pump was controlled, adjusting the water flow rate. Consequently, the water flow rate sensor reading was changed, and the dashboard was updated successfully within 8 seconds, as shown in **Figure (6)**. Finally, the temperature was adjusted simultaneously to test the alert. Alert was launched on approaching the forging temperature at 227°C with a margin of 23°C. The second trial was positive, after considering coding issues and running the modified version of the code, with UV readings of nearly 255 nm.

Figure (6): Timestamp of each data upload



Nuclear Station - Data Storage

	Time Stamp	UV	Temperature	Flow Rate
2	02/12/2024 11:37:46	245	65	0.9
3	02/12/2024 11:37:54	243	92	1.1
4	02/12/2024 11:38:01	256	117	1.2
5	02/12/2024 11:38:08	252	130	1.9
6	02/12/2024 11:39:07	248	143	2.2
7	02/12/2024 11:39:15	253	152	2
8	02/12/2024 11:39:23	258	155	3.3
9	02/12/2024 11:39:29	255	159	2.9
10	02/12/2024 11:39:37	253	162	3.2
11	02/12/2024 11:39:43	265	163	4.3
12	02/12/2024 11:39:50	272	163	4.8
13	02/12/2024 11:39:57	264	161	4.5
14	02/12/2024 11:40:04	262	157	5.5

The figure shows the google spreadsheet, storing sensors' readings and providing history with timestamps and time difference between uploads

Conclusion

After analyzing the project prototype and test results, some major conclusions were made. First, the flow rate of the coolant water inversely correlates with the rate of change of temperature of the core, since more water coming to the core means more heat absorption, decreasing the temperature of the core. Consequently, if the flow rate is high, the temperature will generally have a low rate of change or even be negative, whereas a low flow rate means a higher rate of change. Second, thermocouple type K is the most suitable for the core temperature due to its sensitivity to small changes in temperature, which triggers the alarm as soon as the temperature reaches the threating limit. Third, the GUVA S12SD target wavelength includes the UVC wavelength, which fits the nature of UV radiation in nuclear reactors. Fourth, the mapping indexes of the UV sensor are inverse to the wavelength, requiring the code to account for it and reverse the beginning and end to get accurate results. Finally, using the flow rate to detect the working state of the pump is an efficient and cheaper way to check the working state of the pump compared to the vibration and rotation monitoring in the Olkiluoto Nuclear Power Plant.

Recommendations

Finally, the project has come to an end. Here are some recommendations to improve this project and ensure that it works more efficiently in the future. These recommendations are as follows:

- Monitoring other types of radiation, like alpha, beta, and gamma, to enhance protection against all kinds of leaks.
- Applying machine learning and Artificial Intelligence to radiation sensor reading to distinguish true radiation signals from noise and to cancel the noise more efficiently using destructive interference.
- Using optical fibers to connect the sensors and microcontroller, as they offer speeds 10 to 20 times faster than normal cables, enhancing the Real-Time presentation of the data.
- Coating the sensors with radiation-absorbing materials like lead to enhance accuracy in high radiation zones.
- Developing and designing a website with custom-built data visualization features and user interface, instead of pre-made online visualization tools.
- Implementing the monitoring system in the upcoming Dabaa Powerplant

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